

The Manual Is Not the Mind: On Underdetermination of Origins from Operational Descriptions

Abstract.

A system's operational description—its “user manual”—can be internally consistent and complete with respect to usage, yet underdetermined with respect to its generative origin. This essay argues that when multiple design processes yield indistinguishable operational structures, no internal analysis can uniquely reconstruct the selection mechanism that produced them. The result is not a gap of knowledge but a structural limit: **inverse non-uniqueness**. We formalize this limit, connect it to familiar results in science and engineering, and clarify its implications for models that presume internal closure.

1. The Central Claim

Consider a system S together with an operational description M that specifies how to use S : inputs, outputs, constraints, and procedures. Suppose M is:

- **Operationally complete:** any admissible use of S can be carried out using M .
- **Internally consistent:** the rules in M do not contradict each other.

The claim is:

*Even if M is complete and consistent for operation, it can be **underdetermined** with respect to the **generative origin** of S . That is, there may exist multiple distinct design processes $\{D_i\}$ such that each D_i yields a system whose operational behavior is captured by the same M .*

Formally, let $\mathcal{D}$ be a space of design processes and $\mathcal{O}$ the induced operational behaviors. The forward map

$$\Phi : \mathcal{D} \rightarrow \mathcal{O}$$

need not be injective. If $\Phi(D_1) = \Phi(D_2) = M$ with $D_1 \neq D_2$, then the inverse problem $\Phi^{-1}(M)$ is **set-valued**. The origin is not uniquely recoverable.

2. From Completeness to Underdetermination

It is tempting to equate “complete manual” with “complete explanation.” This is a category error.

- **Operational completeness** concerns *what can be done* with the system.
- **Generative completeness** concerns *how the system came to be*.

The former does not imply the latter. A manual can encode every permissible action while remaining silent about the **selection path** that fixed one design among many viable alternatives. The missing element is not an omitted parameter; it is the **non-uniqueness of the inverse mapping**.

3. Inverse Non-Uniqueness as a Structural Limit

The phenomenon is familiar across disciplines:

- **Inverse problems:** Distinct parameter fields can produce identical observations (non-uniqueness of inverse solutions).
- **Non-convex optimization:** Multiple minima yield comparable performance; different training trajectories lead to models with indistinguishable outputs.
- **Dynamical systems:** In regimes without uniqueness guarantees (e.g., violations of Lipschitz conditions), multiple trajectories satisfy the same local laws.

In each case, forward adequacy (the model reproduces observations) does not entail inverse identifiability (the underlying cause is uniquely determined). The limit is structural, not merely epistemic.

4. Selection Without Reconstruction

If multiple D_i map to the same M , then any attempt to “recover the designer” from M alone faces a dilemma:

1. **Assume a unique inverse:** contradicts the existence of multiple D_i .
2. **Augment the model with additional internal variables:** risks circularity if those variables are not independently constrained by M .
3. **Accept set-valued origins:** concede that the selection among $\{D_i\}$ is not derivable from M .

The third option is the only consistent one without introducing ad hoc structure. It implies that the **selection mechanism**—the rule that picks one D_i from the admissible set—is not recoverable from the operational description alone.

5. Relation to External Selection (TNA Perspective)

Within frameworks like the Theory of Non-Admissible Derivability (TNA), this manifests as follows:

- Let N_0 denote the internal dynamics and admissible descriptions (the “manual”).
- Let N_1 denote the selection operator over admissible trajectories or designs.

When the system is in a regime where multiple admissible outcomes exist (e.g., $\Delta > 1$), no operator defined purely within N_0 can uniquely select among them **without reintroducing the very indeterminacy it aims to resolve**. The necessity of N_1 is therefore not a metaphysical claim about “outside the universe,” but a **statement about non-derivability within the internal description**.

6. Objection: “Add the Missing Variables”

A common response is that the manual is simply incomplete; add the right hidden variables and the inverse becomes unique.

This move succeeds only if the added variables are:

- **Independently observable or constrained**, and
- **Sufficient to restore injectivity** of Φ .

In many real systems, neither condition holds. One can always *postulate* variables that encode the selection, but unless they are anchored by independent evidence, this is a re-labeling of the problem, not a solution. The inverse remains underdetermined.

7. Implications

1. Limits of Reverse Engineering

Perfect operational knowledge does not guarantee reconstruction of origin.

Engineering practice implicitly acknowledges this via design spaces, trade-offs, and non-unique solutions.

2. **Cautious in Causal Claims**

Observational equivalence undermines strong claims about unique causes. Competing generative stories may be indistinguishable at the operational level.

3. **Modeling Discipline**

Distinguish clearly between *fit to behavior* and *identifiability of causes*. The former does not license the latter.

4. **Role of Selection Principles**

Where non-uniqueness is intrinsic, selection must be treated explicitly—either as an external operator, a prior, or a decision rule—rather than presumed derivable from within the operational model.

8. Conclusion

A manual can be flawless for use and still be silent about origin. When multiple generative paths converge to the same operational structure, the inverse problem is fundamentally non-unique. This is not a temporary gap to be closed by more data or clever parameterization; it is a structural feature of the mapping from design to behavior. Recognizing this limit clarifies where explanation ends and where **selection**—as an independent component—must begin.